

EXPERIMENT 6

6.1.1 Incremented Date

ALGORITHM

Step 1: Start the program.

Step 2: Input day (DD).

Step 3: Input month (MM).

Step 4: Input year (YYYY).

Step 5: Check if year ≤ 0 .

If yes, print "Invalid Date" and go to Step 20.

Step 6: Check if month < 1 or month > 12 .

If yes, print "Invalid Date" and go to Step 20.

Step 7: Determine if the year is a leap year using the condition:

If (year % 400 == 0) OR (year % 4 == 0 AND year % 100 $\neq$ 0),

then leap_year = True,

else leap_year = False.

Step 8: Set maximum days in the month:

If month is 1, 3, 5, 7, 8, 10, or 12 $\rightarrow$ max_days = 31.

Step 9: Else if month is 4, 6, 9, or 11 $\rightarrow$ max_days = 30.

Step 10: Else if month is 2:

If leap_year is True $\rightarrow$ max_days = 29.

Else $\rightarrow$ max_days = 28.

Step 11: Check if day < 1 or day $>$ max_days.

If yes, print "Invalid Date" and go to Step 20.

Step 12: If day $<$ max_days, then

new_day = day + 1

new_month = month

new_year = year

Step 13: Else if day == max_days and month $\neq$ 12, then

new_day = 1

new_month = month + 1

new_year = year

Step 14: Else if day == 31 and month == 12, then

new_day = 1

new_month = 1

new_year = year + 1

Step 15: Display the incremented date in the format

DD-MM-YYYY

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Step 16: End the program.

PYTHON CODE

```
day = int(input())
```

```
month = int(input())
```

```
year = int(input())
```

```
def is_leap_year(year):
```

```
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
```

```
        return True
```

```
    return False
```

```
if year <= 0:
```

```
    print("Invalid Date")
```

```
elif month < 1 or month > 12:
```

```
    print("Invalid Date")
```

```
else:
```

```
    if month in [1, 3, 5, 7, 8, 10, 12]:
```

```
        max_days = 31
```

```
    elif month in [4, 6, 9, 11]:
```

```
        max_days = 30
```

```
    elif month == 2:
```

```
        if is_leap_year(year):
```

```
            max_days = 29
```

```
    else:
```

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```
max_days = 28
```

```
if day < 1 or day > max_days:
```

```
    print("Invalid Date")
```

```
else:
```

```
    day += 1
```

```
if day > max_days:
```

```
    day = 1
```

```
    month += 1
```

```
if month > 12:
```

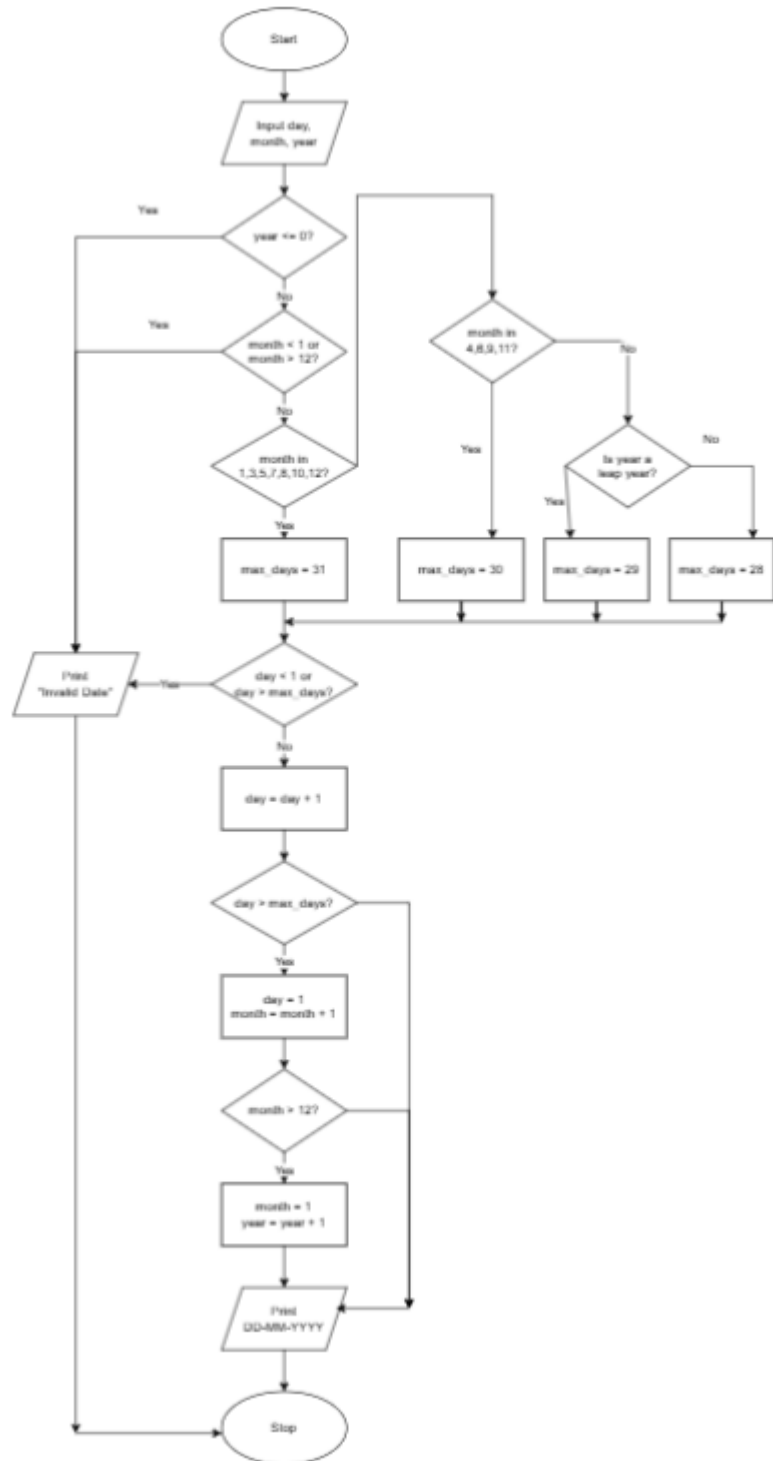
```
    month = 1
```

```
    year += 1
```

```
print(f"{day:02d}-{month:02d}-{year}")
```

FLOWCHART

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EXCECUTION

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CODETANTRA

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karthiyanigal@tantra.ac.in · Support · Logout

6.1.1 Incremented Date

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

- Valid Month:** 1 to 12
- Valid Day:** Depends on the month and year
 - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
 - April (4), June (6), September (9), November (11): 1 to 30 days
 - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
- Valid Year:** Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format: `dd-mm-yyyy`
- If the date is invalid, print: "Invalid Date".

Sample Test Cases

nextDate.py

```
1 day = int(input())
2 month = int(input())
3 year = int(input())
4
5 def is_leap_year(year):
6     if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
7         return True
8     return False
9
10 def days_in_month(month, year):
11     if month in [1, 3, 5, 7, 8, 10, 12]:
12         return 31
13     elif month in [4, 6, 9, 11]:
14         return 30
15     elif month == 2:
16         if is_leap_year(year):
17             return 29
18         else:
19             return 28
20     else:
21         return 0
22
23 if year <= 0 or month < 1 or month > 12:
24     print("Invalid Date")
25 else:
26     max_days = days_in_month(month, year)
27     if day < 1 or day > max_days:
28         print("Invalid Date")
29     else:
```

Execution Time: 0.025 s

Maximum Time: 0.100 s

32.44 ms

100.00 ms

5 out of 5 shown test cases passed

5 out of 5 hidden test cases passed

Test case 1

Expected output

Actual output

20-09-2024

20-09-2024

Terminal

Test cases